

3964. Minimum Lights to Illuminate a Road

Solved ●

Medium



Hint

You are given an integer array `lights` of length `n`, representing positions `0` through `n - 1` on a road.

For each position `i`:

- If `lights[i] = v`, where `v > 0`, there is a working bulb at position `i` that **illuminates** every position from `max(0, i - v)` to `min(n - 1, i + v)`, inclusive.
- If `lights[i] = 0`, there is no working bulb at position `i`.

A position is **visible** if it is illuminated by **at least** one working bulb.

You may install **additional** bulbs at **any** positions. Each additional bulb installed at position `j` **illuminates** positions from `max(0, j - 1)` to `min(n - 1, j + 1)`, inclusive.

Return the minimum number of additional bulbs required to make **every** position on the road visible.

Example 1:

Input: `lights = [0,0,0,0]`

Output: 2

Explanation:

One optimal placement is:

- Install an additional bulb at position 1, illuminating positions `[0, 1, 2]`.
- Install an additional bulb at position 3, illuminating positions `[2, 3]`.

Therefore, the minimum number of additional bulbs required is 2.

Example 2:

Input: `lights = [0,0,0,2,0]`

Output: 1

Explanation:

- Since `lights[3] = 2`, the working bulb at position 3 illuminates positions `[1, 2, 3, 4]`.
- Installing an additional bulb at position 1 illuminates positions `[0, 1, 2]`, making every position visible.
- Therefore, the minimum number of additional bulbs required is 1.

Constraints:

- `1 <= n == lights.length <= 105`
- `0 <= lights[i] <= n`

Seen this question in a real interview before? 1/6

Yes No

Accepted 23,580/62.5K | Acceptance Rate 37.7%

Topics



Hint 1



Hint 2



Hint 3



Discussion (51)

